

Why L-Compute Eliminates Mamba Projection-SRAM Bank Conflicts

Exact x-field packet for one Nemotron 3 group: X_STATE requests 8 heads $\times$ 4 adjacent P-lanes = 32 values from 16 single-port banks

(a) Baseline: Row-Major Matrix Writeback

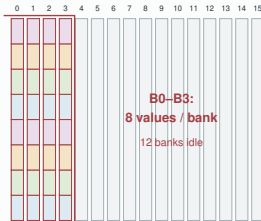


bank = p because each 64-value head row starts again at bank 0

Bank selected for each logical value

	P-lane			
	p0	p1	p2	p3
h0	B0	B1	B2	B3
h1	B0	B1	B2	B3
h2	B0	B1	B2	B3
h3	B0	B1	B2	B3
h4	B0	B1	B2	B3
h5	B0	B1	B2	B3
h6	B0	B1	B2	B3
h7	B0	B1	B2	B3

Packet occupancy in 16 banks



Correct packet, but the four hot banks serialize X_STATE reads

(b) Proposed: L_SCATTER_M Skew-on-Write

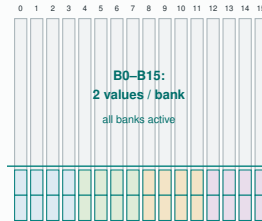


shift each head row by 4 banks the logical tensor and P-lane order do not change

Bank selected for each logical value

	P-lane			
	p0	p1	p2	p3
h0	B0	B1	B2	B3
h1	B4	B5	B6	B7
h2	B8	B9	B10	B11
h3	B12	B13	B14	B15
h4	B0	B1	B2	B3
h5	B4	B5	B6	B7
h6	B8	B9	B10	B11
h7	B12	B13	B14	B15

Packet occupancy in 16 banks



Same packet, balanced across all banks at the 32/16 lower bound